

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [104, 97, 97, 92, 79, 88, 87, 83, 84, 76]

Tap amplitudes (pixels): [203.8062805705457, 204.61182761512103, 193.0932417253385, 183.91846019364124, 192.77447963877376, 186.46179233290664, 189.73665961010275, 174.45343218177166, 171.42053552593984, 182.8578683021324, 166.23477373883]

GPT-5

MDS-UPDRS 3.4 Finger Tapping Scoring Analysis

Given data:

- Number of taps recorded: 11 for a 10-tap task (common when start/end events are included; we'll assess the first 10 inter-tap intervals and all 11 amplitudes).
- Tap periods (frames) for 10 intervals: [104, 97, 97, 92, 79, 88, 87, 83, 84, 76]
- Tap amplitudes (pixels) for 11 taps: [203.8, 204.6, 193.1, 183.9, 192.8, 186.5, 189.7, 174.5, 171.4, 182.9, 166.2] (rounded for readability)

Interpretation approach:

- Speed: assess slowing via trend in periods (lower frames = faster; higher = slower). Moderate/marked slowing would manifest as consistently long periods or progressive lengthening. Here, periods generally decrease slightly (faster), not slower. No evident slowing.
- Rhythm interruptions/hesitations: look for outliers or abrupt deviations in period suggesting hesitations/freezes. The sequence is fairly smooth, with small variability (104 → 97 → 97 → 92 → 79 → 88 → 87 → 83 → 84 → 76). No long arrest; no >5 interruptions; no clear 3–5 discrete hesitations. Minor variability only.
- Amplitude decrement:
 - Start high (≈204–205) and overall trends downward toward ≈166 by the 11th tap.
 - Compare early taps to mid and late:
 - Early (taps 1–2): ~204–205
 - Mid (taps 5–7): ~193, 186, 190
 - Late (taps 9–11): ~171, 183, 166

- Pattern: There is a general decrement noticeable by mid sequence (tap 3 ~193, tap 4 ~184), partial rebounds occur (e.g., tap 5 ~193, tap 10 ~183), but the envelope declines. This is consistent with a decrement beginning around early–mid taps, not just at the very end. Importantly, the “3” criterion specifies decrement starting after the 1st tap; here the amplitude drops from ~205 to ~193 by tap 3 and ~184 by tap 4—this qualifies as decrement starting soon after initiation, though not strictly monotonic.

Mapping to criteria:

- 0 (Normal): Not applicable—there is a discernible amplitude decrement.
- 1 (Slight): Would fit if decrement occurs only near the end, or if there are just 1–2 hesitations or slight slowing. The decrement here appears earlier than “near the end.”
- 2 (Mild): Matches if amplitude decrements midway through the 10 taps, with no significant hesitations or slowing. This aligns well: decrement is evident by taps 3–5 (mid-sequence), rhythm is fairly regular, and no significant slowing.
- 3 (Moderate): Would require >5 interruptions or a longer arrest, moderate slowing, or decrement starting right after the 1st tap. While the decrement starts early (after tap 1–2), the absence of interruptions/freezes and lack of slowing make “3” less appropriate unless we strictly enforce “decrement starting after the 1st tap.” Given the non-monotonicity and relatively preserved rhythm/speed, the overall impairment seems closer to mild than moderate.
- 4 (Severe): Not supported.

Conclusion:

- The strongest evidence is a clear amplitude decrement becoming apparent by the mid portion of the sequence, with preserved speed and rhythm and no arrests.

Final score: 2
